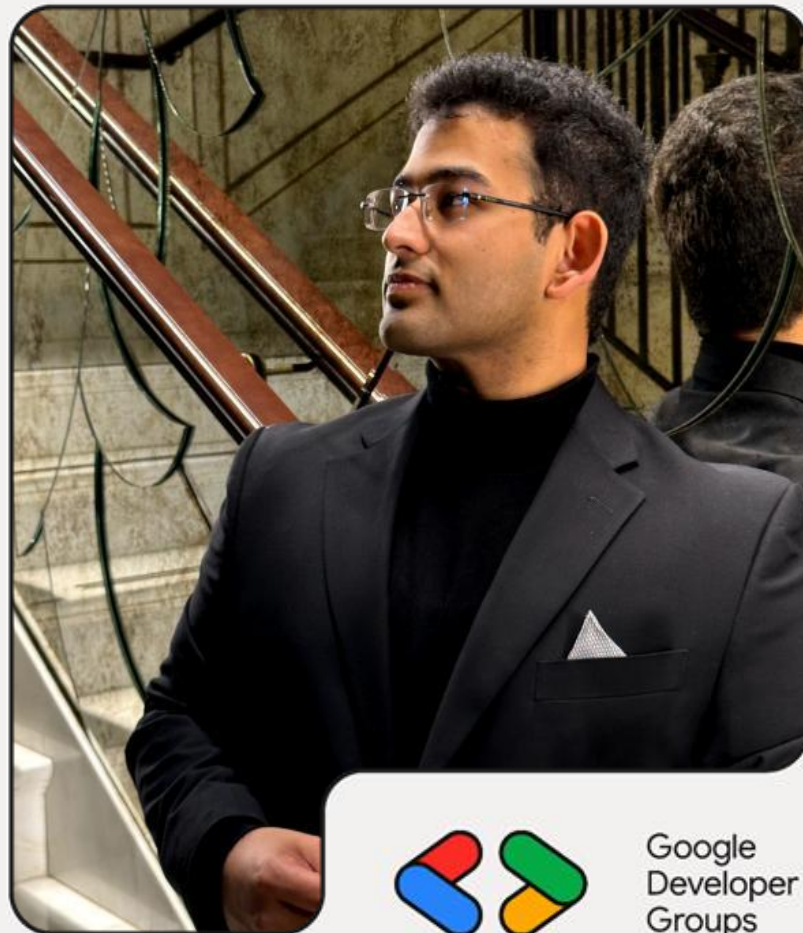




Creating Multi-Agent Workflows with Gemini Enterprise & ADK

Rohan Mitra

ML Engineer @ Bayut | Dubizzle
Research Assistant for ML @ CUD
Organizer & Speaker GDG Sharjah



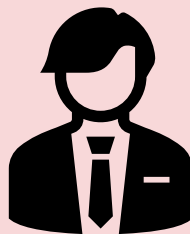
Google
Developer
Groups

LLMs vs Agentic AI



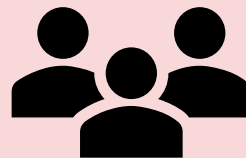
LLM

- Reasoning and knowledge, but can't take actions
- Stateless
- Passive



Agent

- Body = tools
- Job Desc = instructions & goals
- Autonomous
- Stateful
- Take actions till job is done



Multi-Agent

- Talented and focused individuals
- Get tasks to help achieve goal
- Focused
- Modular, reliable, debug-able

Using LLMs & Agents daily



Gemini Enterprise

- Formerly AgentSpace
- Secure: Data not used to improve models
- Easy integration to data sources
- Use Google-made or partner-made agents
- Create no-code agents & workflows
 - Make your own via prompts or manually
 - Suitable for non-tech teams
- Integrate custom ADK agents and workflows – fun for devs too ;)



New chat

Search

Starred

Library

Agents

Create agent

Pinned

Deep Research

Idea Generation

Tips to get started



Connect your data

Ask a question

Try an agent

Personalize your experience

Invite team members

Chats

Hypermarket marketing plan

- [Dubai, Dubai, AE](#)
From your IP address
[Learn more](#)

Free trial

◆ Hello, Rohan Mitra

Gemini for Rohan Mitra's Team

Ask anything, search your data, @mention or /tools



4 of 12

◆ Auto



Discover what's possible

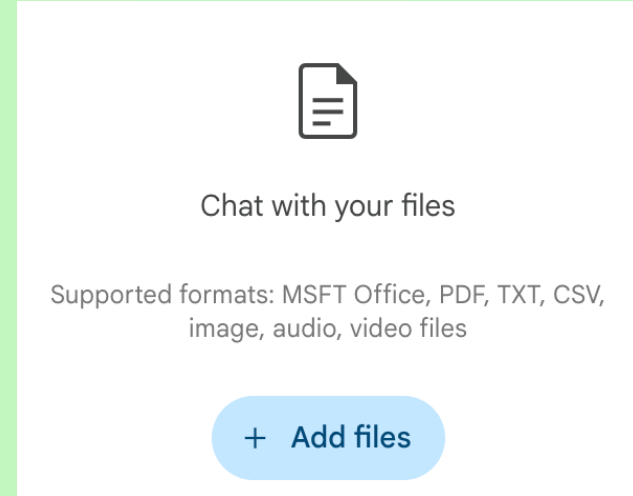
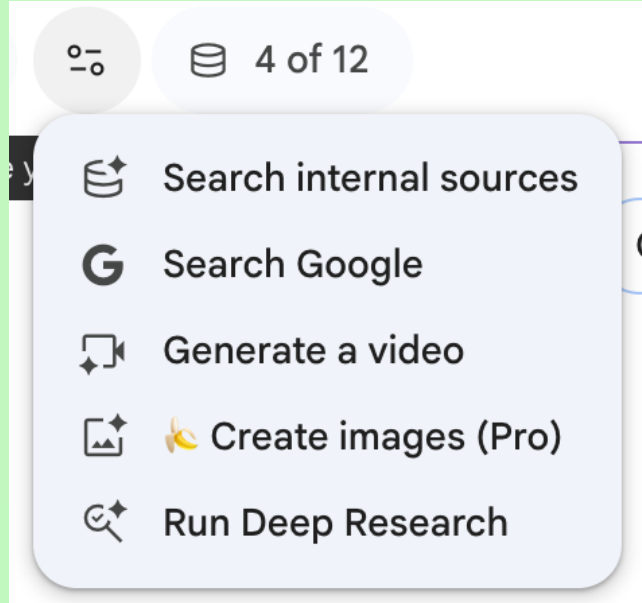
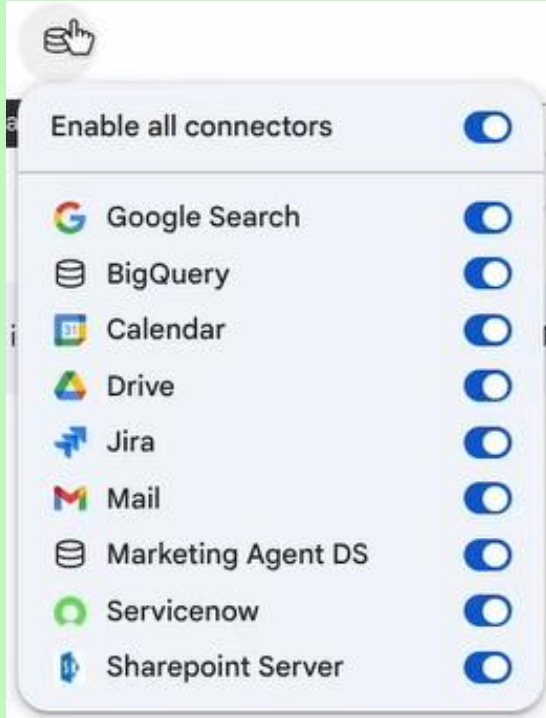
Go off-script

Spark a breakthrough

Start my day

Gemini Enterprise

Connect with data sources – prebuilt, uploaded files, MCP servers



Gemini Enterprise

Agent Designer – via prompts, drag & drop, or ADK agents!

 Gemini Enterprise

Auto ▾

 **Agent Designer**

Let me know if you wish to make any changes to your agent 'Campaigns Agent'.

 Ask to make changes

+

📄

▶

 **Campaigns Agent** ▾ 

Draft

Flow

Details

Preview

↶

↷

Update

⚙️ Campaigns Agent

Coordinates core recurring processes for our seasonal campaigns.

⚙️ Market Research Agent

Identifies trends and performs competitive analysis.

⚙️ GenMedia Agent

Generate copy, images, and video with company's brand guidelines in mind.

⚙️ Comms Agent

Drafts and Distributes team communications on different channels.

⚙️ Inventory Agent

Checks and adjusts the stock of products.

+

📄

▶

⌕

+

–

Why Agent Development Kit (ADK)?

- Gemini Enterprise is the front-end and very high level
- Need more control to build more complex agents and workflows
 - Modularity
 - Reusability
 - State management
- Deploy to custom apps – build features for product
- Build internal tools – Deploy onto Gemini Enterprise

Case Study – G-Stock Brokers

- You are an AI Engineer at G-Stock Brokers
- The traders want to quickly know how a certain stock is performing in a given time period
- They want some market research on latest news + numerical backing
- They should be able to ask for a given stock and receive a report with relevant info and suggestions
- Let's build an AI workflow with Agent Development Kit!
- (And how you can adapt this to your own projects)

Stock Market Analysis Bot

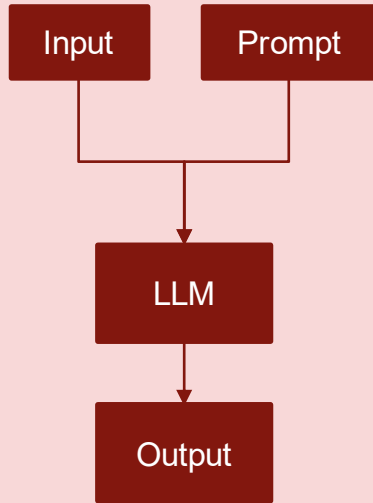
- Take a particular stock and time period, and conduct analysis on how the stock performed
- Search the web for headlines about the stock
- Analyze the market sentiment towards the stock
- Fetch and analyze temporal data (prices & volume of the stock) in that time period
- Collate data, summarize, and identify actionable insights
- Make an organized report

Adaptable Design – Other Use Cases

- Weekly sales reports - Sales
- Daily operations status reports - Operations
- Chatting with your data - General
- Strategic planning - Strategy
- Automate global financial consolidation and budget modeling - Finance

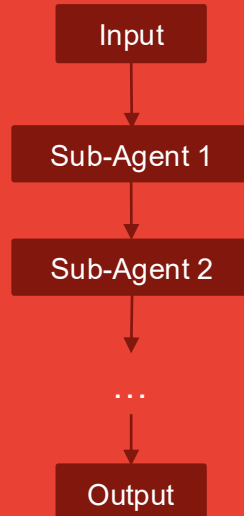
ADK Agents

LLM Agent

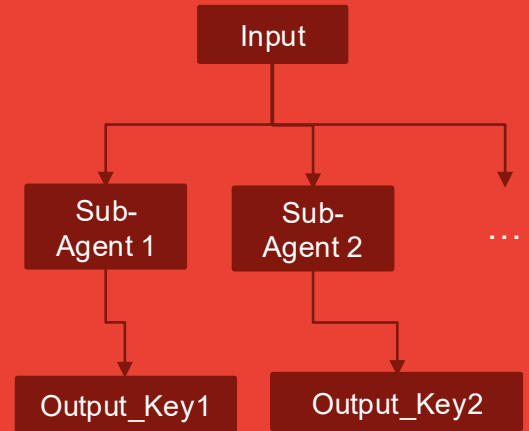


Workflow Agent

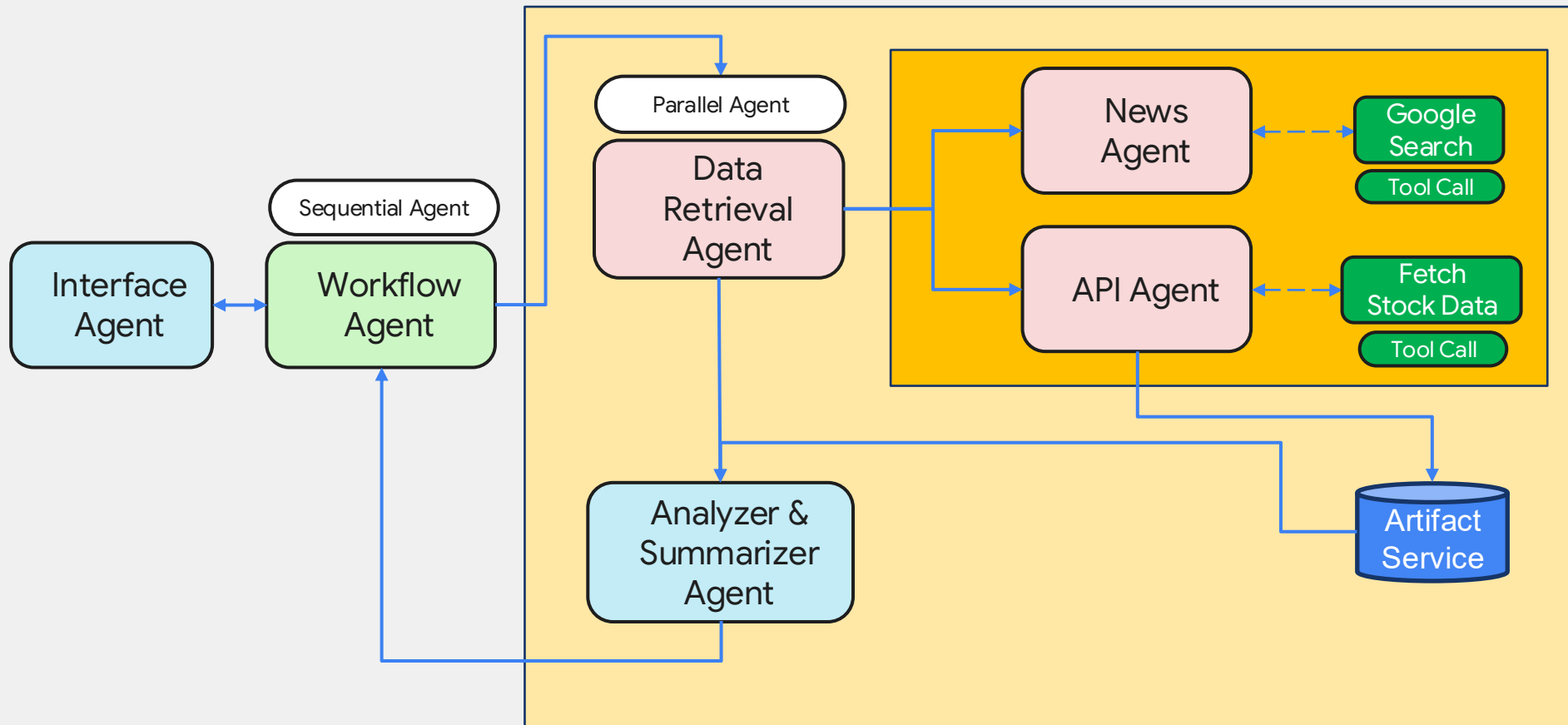
Sequential Agent



Parallel Agent



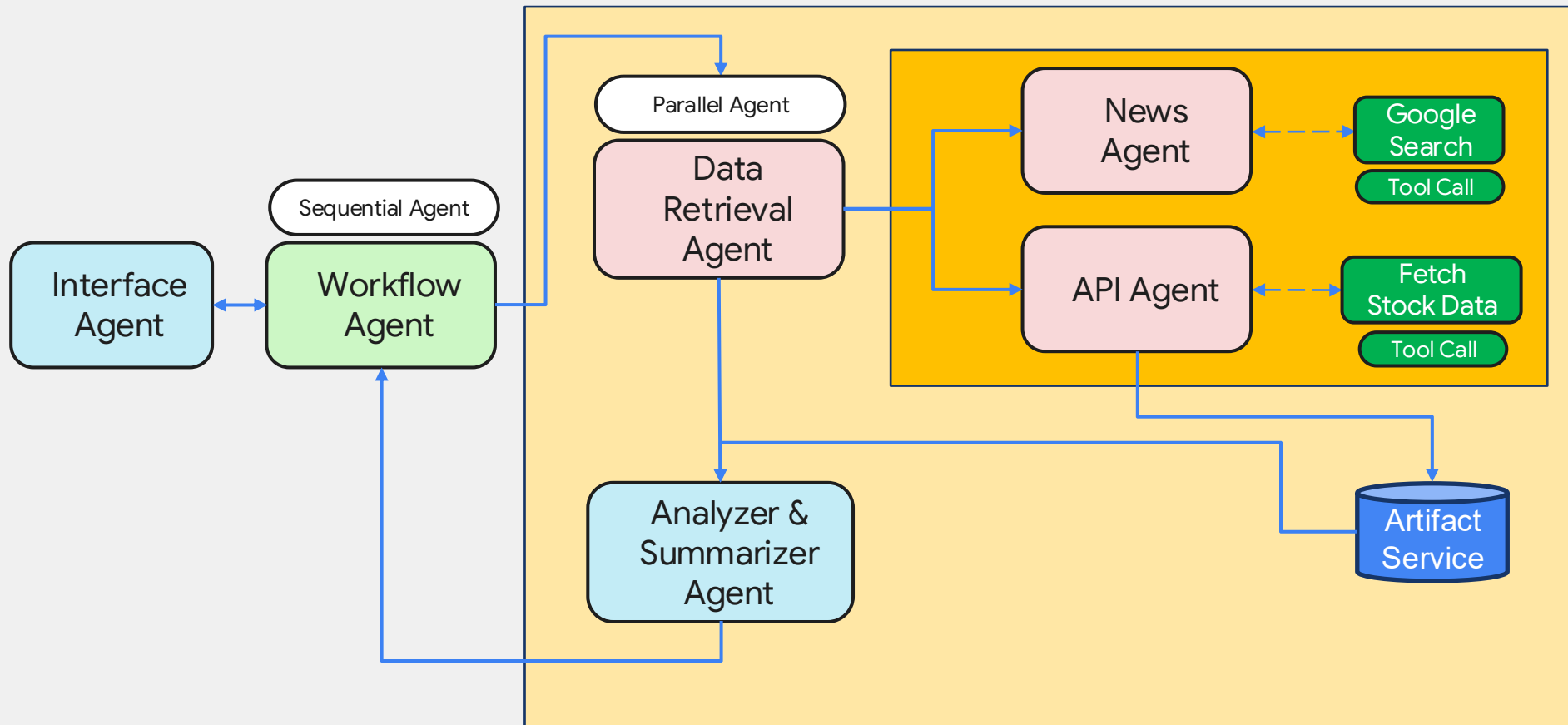
Our Architecture



Interface Agent

- Interacts with the user
- Ask the user for a stock (ticker or name of publicly listed company)
- Ask the user for a date range
- Validate inputs
- Initiate workflow
- Return result to user as message

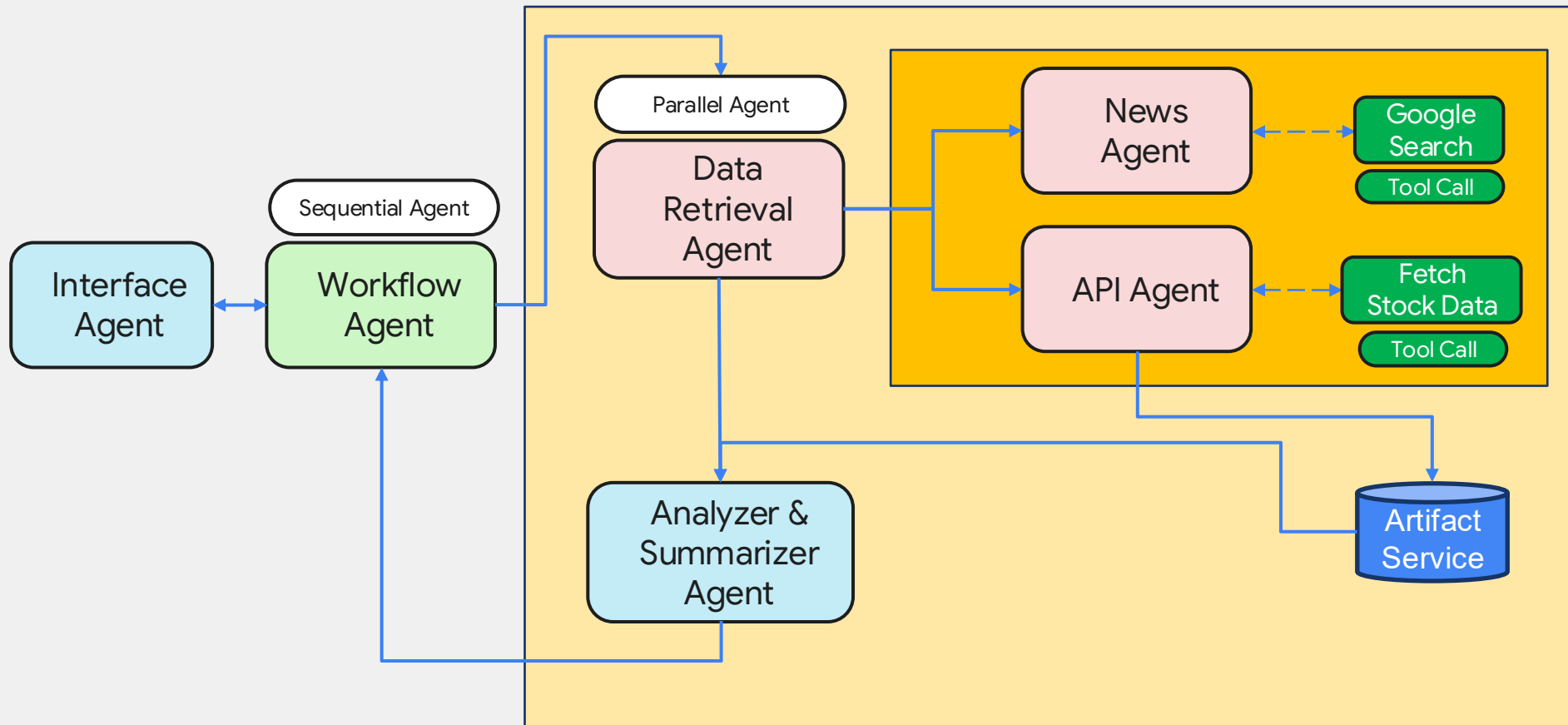
Our Architecture



Workflow Agent

- Sequential agent
- Input text from the Interface agent
 - Eg: “Analyze GOOG stock performance, news sentiment, technical indicators, and provide trading recommendations for the period 2025-08-01 to 2025-08-31.”
- First execute Data Retriever Agent (result stored in output_key)
- Then execute Analyzer & Summarizer Agent (result stored in output_key)
- Once done, return control to Interface agent

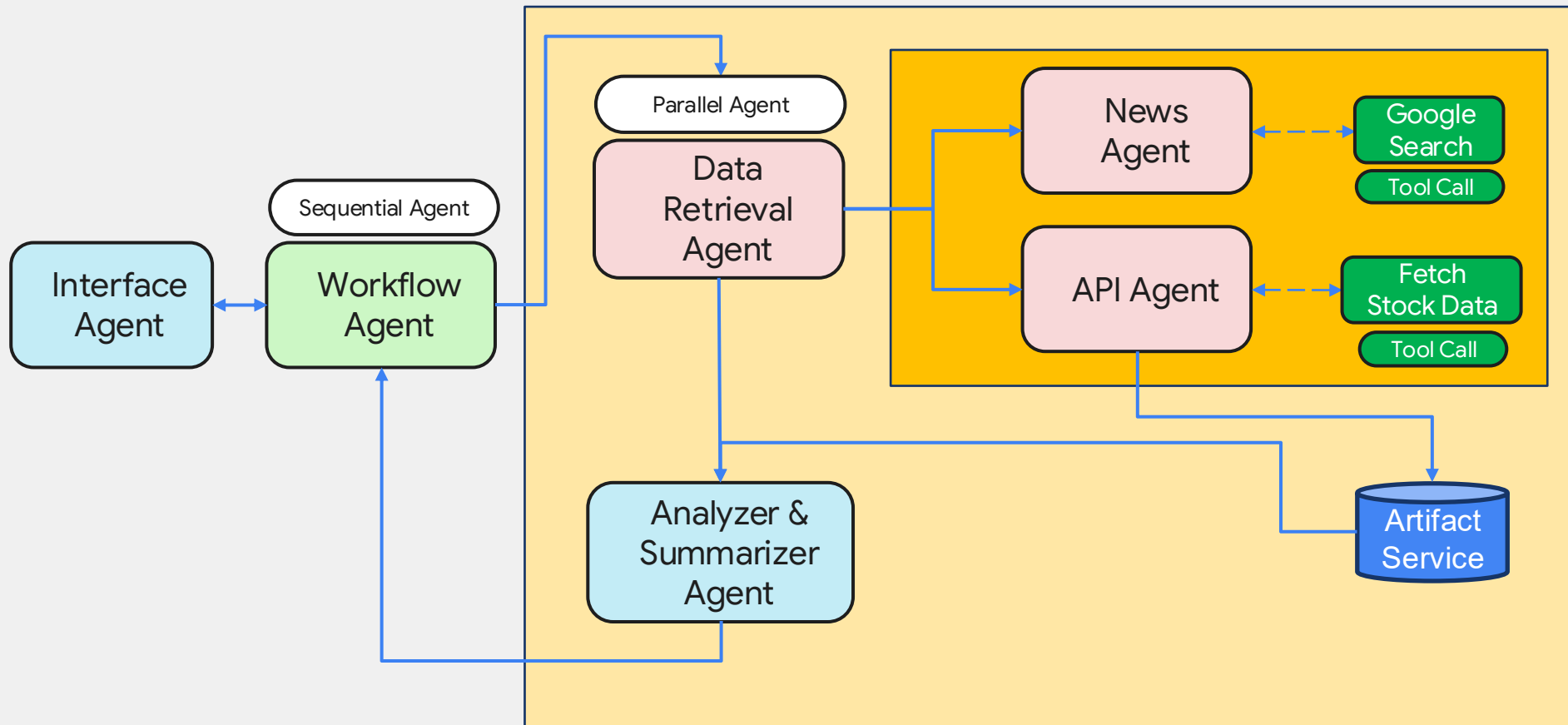
Our Architecture



Data Retrieval Agent

- Parallel agent
- Execute News Agent
- Execute API Agent
- Once done, return control to Workflow agent

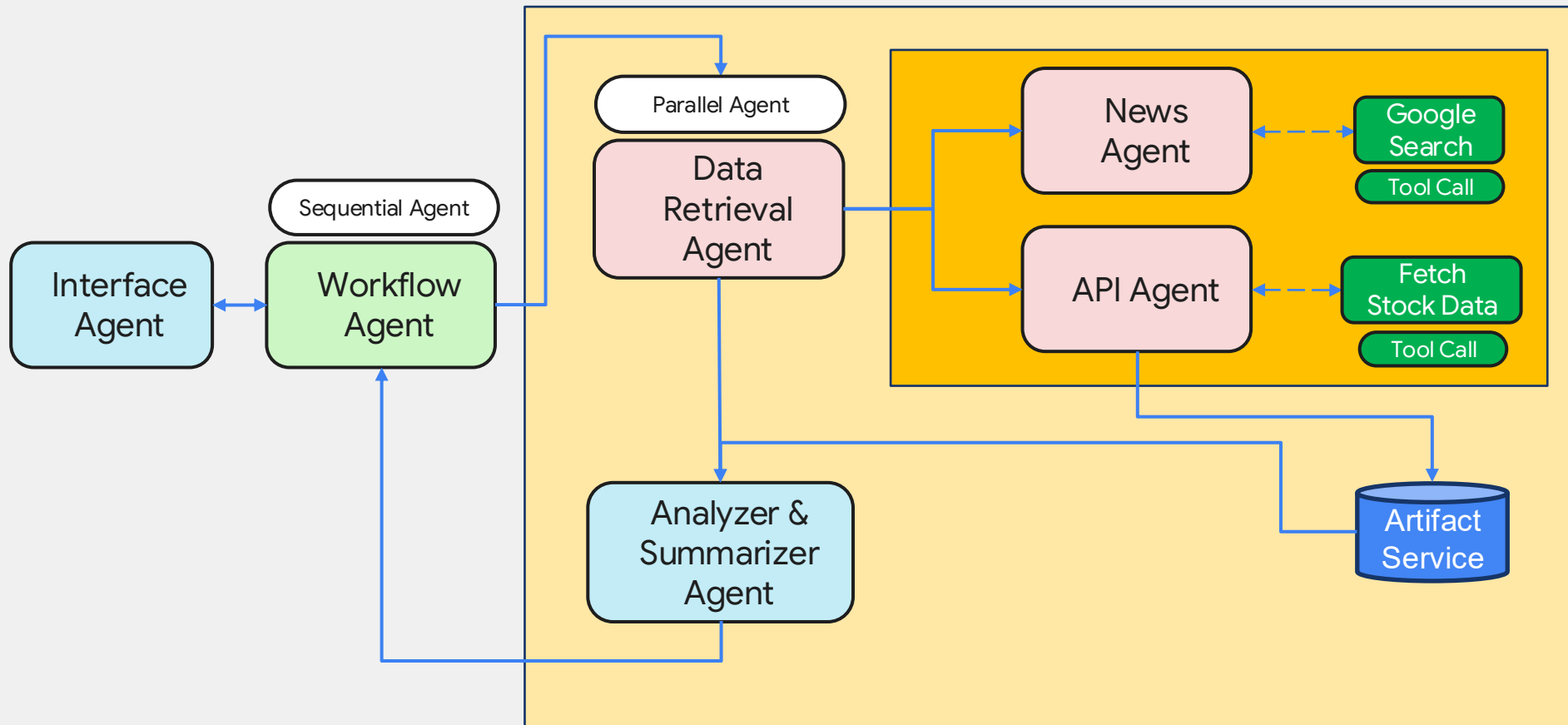
Our Architecture



News Agent

- Performs several Google searches with different search keys – Google Search tool call
- Identifies:
 - Summary of latest news articles & developments of given stock
 - Top 5 most impactful headlines
 - Market sentiment (Positive, Neutral, Negative)
- Saves the results in JSON format in an output key “news_agent_output”

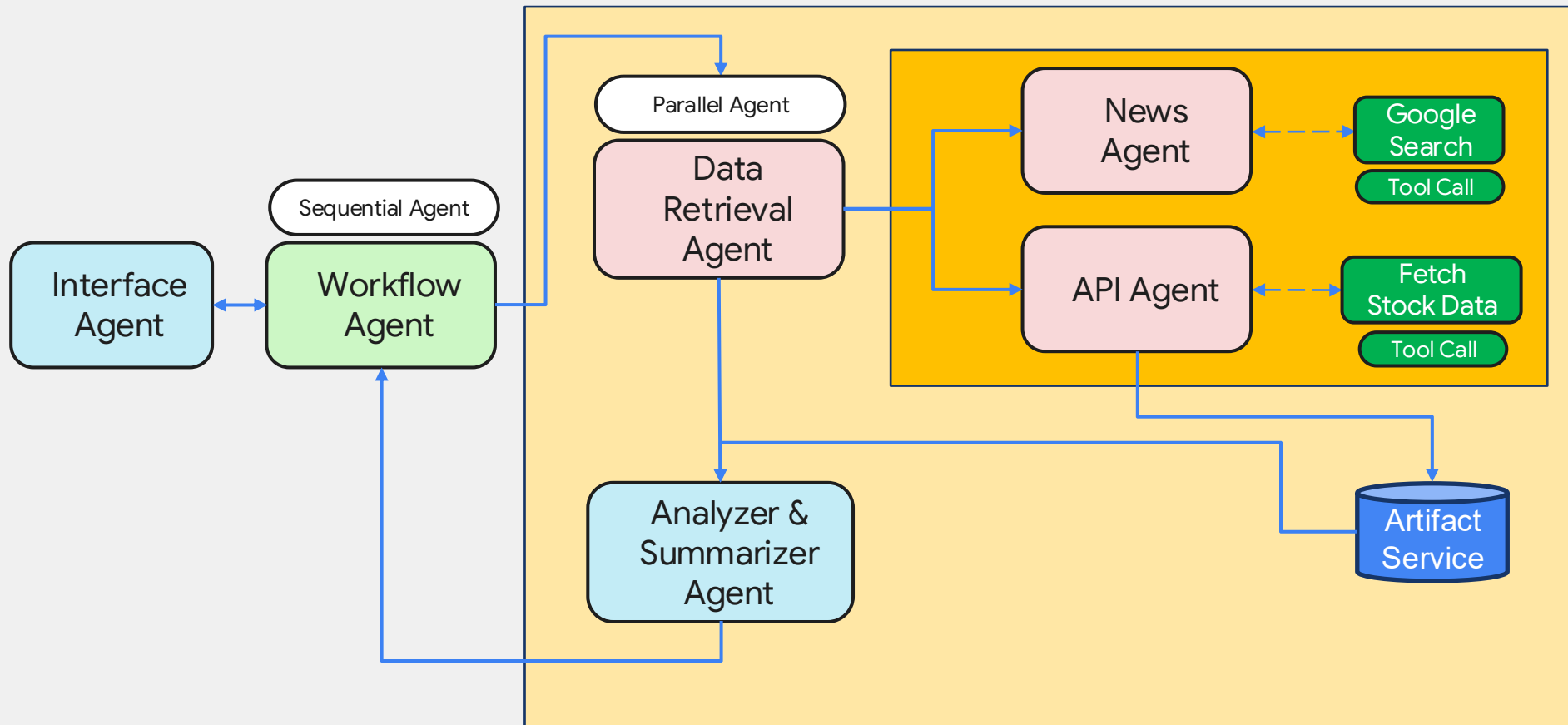
Our Architecture



API Agent

- Uses a tool-call to fetch data for stock from Yahoo finance API (yfinance)
- Downloads the open/close/high/low/volume data (as dataframe) for each date
- Saves the data as a JSON in artifact service
 - **Artifact:** Any large object that is loaded/created
 - Includes files, images, data, etc
 - **Artifact service:** A temporary storage space for the session
 - Easy way to transfer large objects between agents
 - Saves to RAM / GCS / Redis

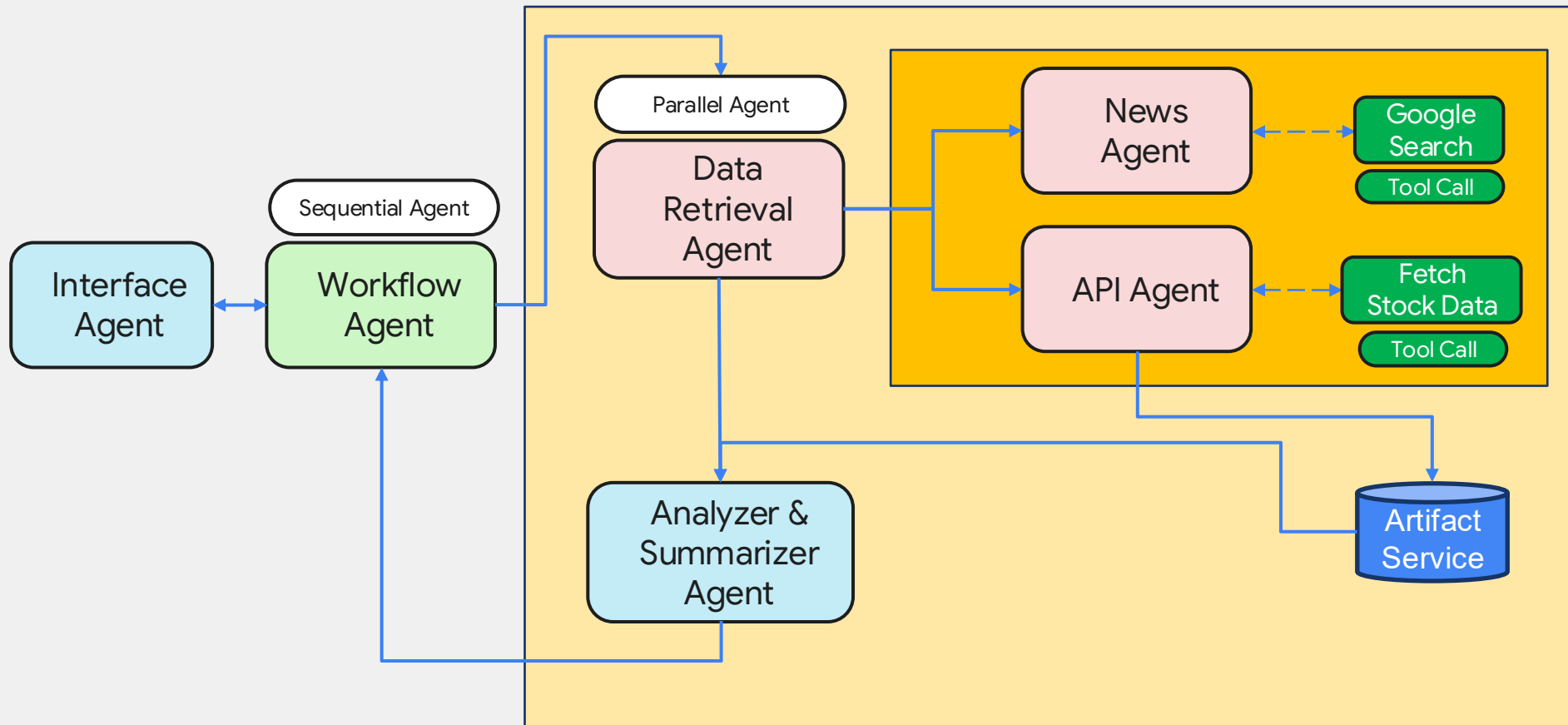
Our Architecture



Analyzer & Summarizer Agent

- Analyze news data (news_agent_output)
- Load and analyze temporal data
 - Load the data from artifacts
- Combined analysis
 - Correlation, timing of news vs stock prices, market reactions to different types of news
- Return summary of analysis to Workflow Agent
 - News, Trends, Combined Insights, Trading Recommendations, Tables

Our Architecture



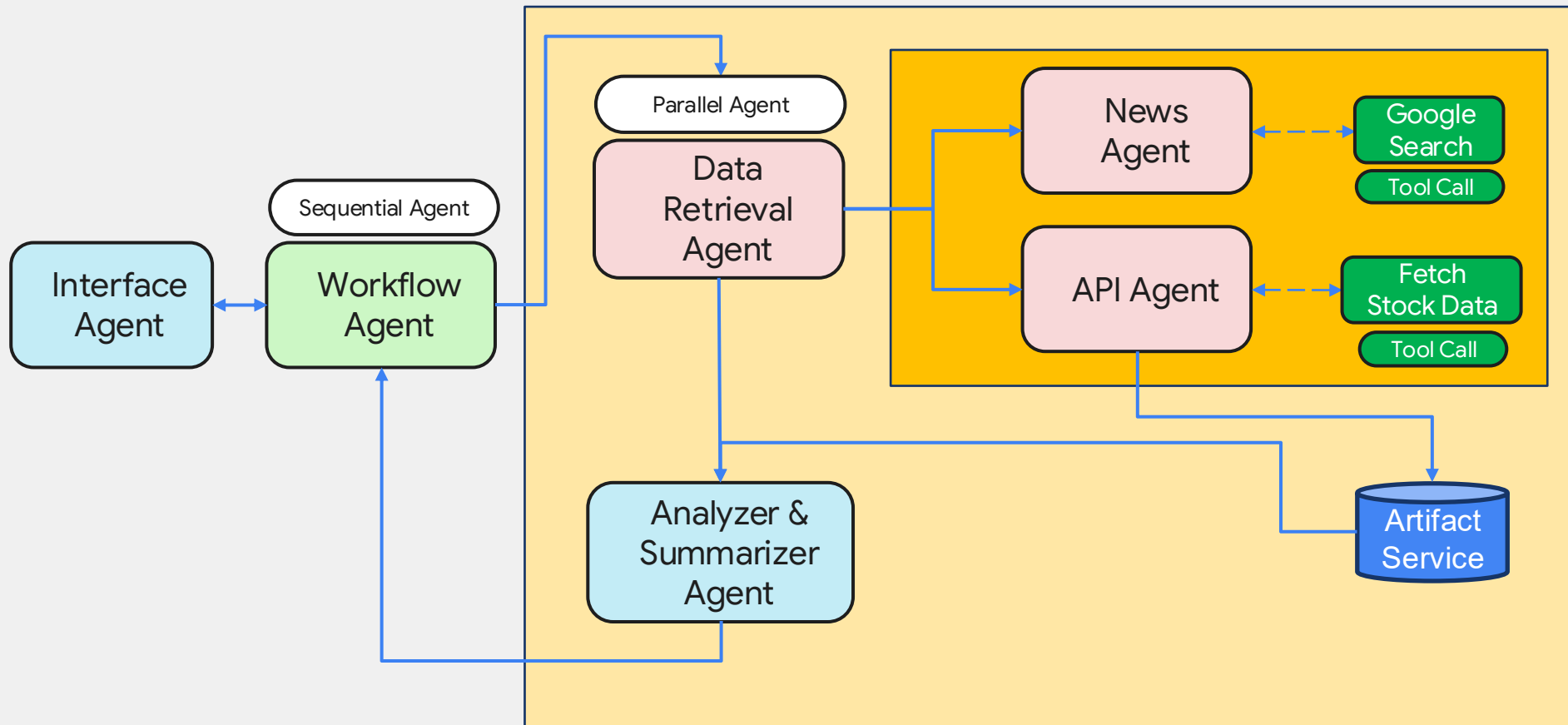
Adapt To Your Use Case

- ADK forces modular architectures
- Plug-and-play
- Update Prompts
- Add / Replace tools & agents

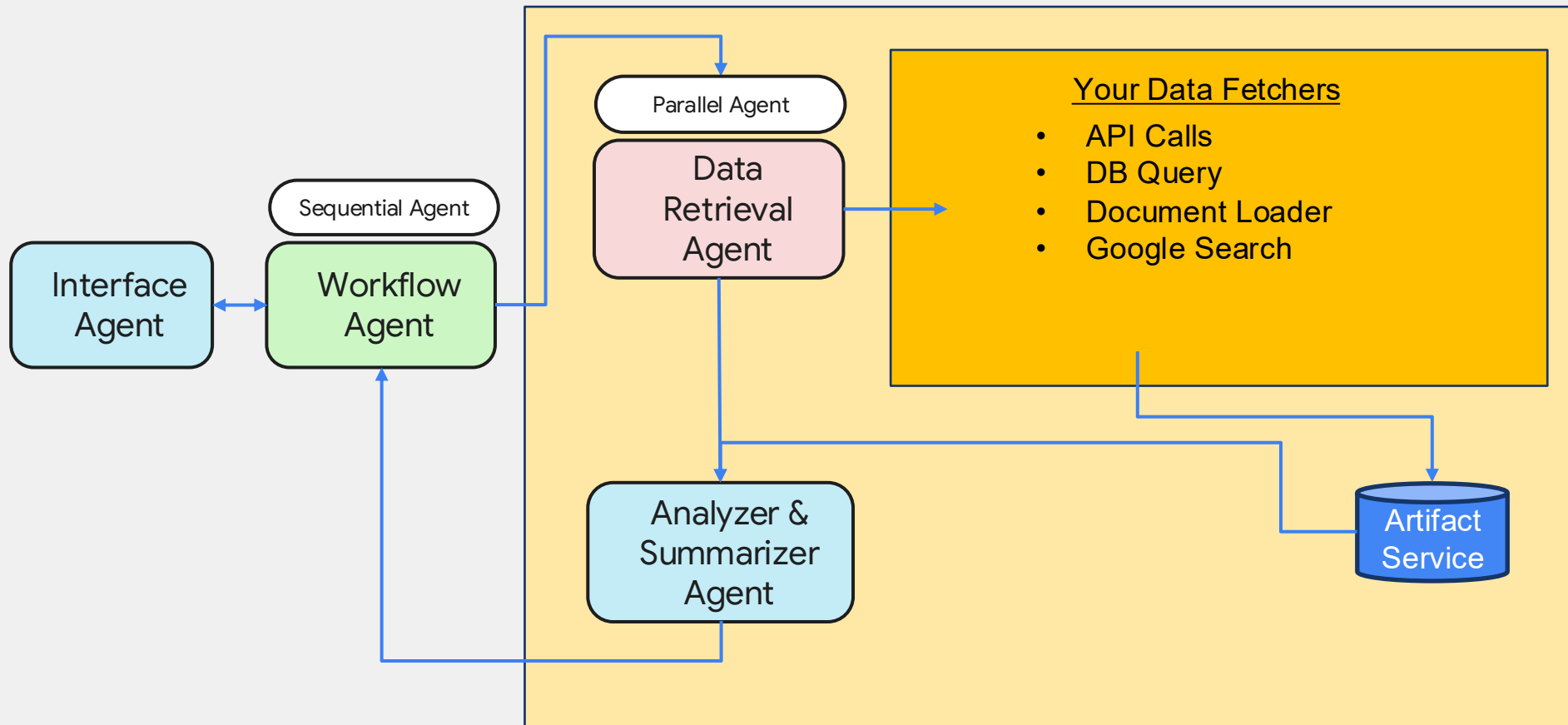
Github repo with code



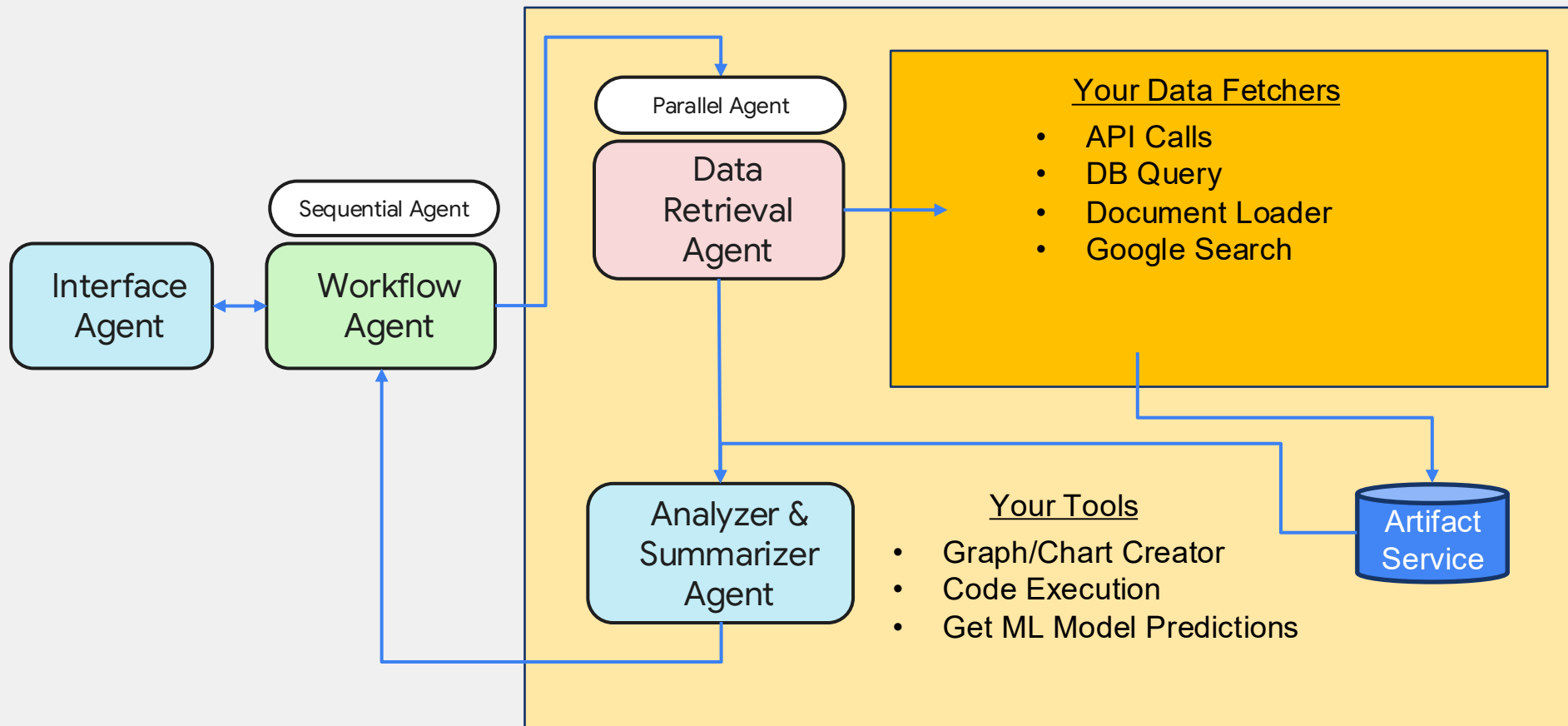
Our Architecture



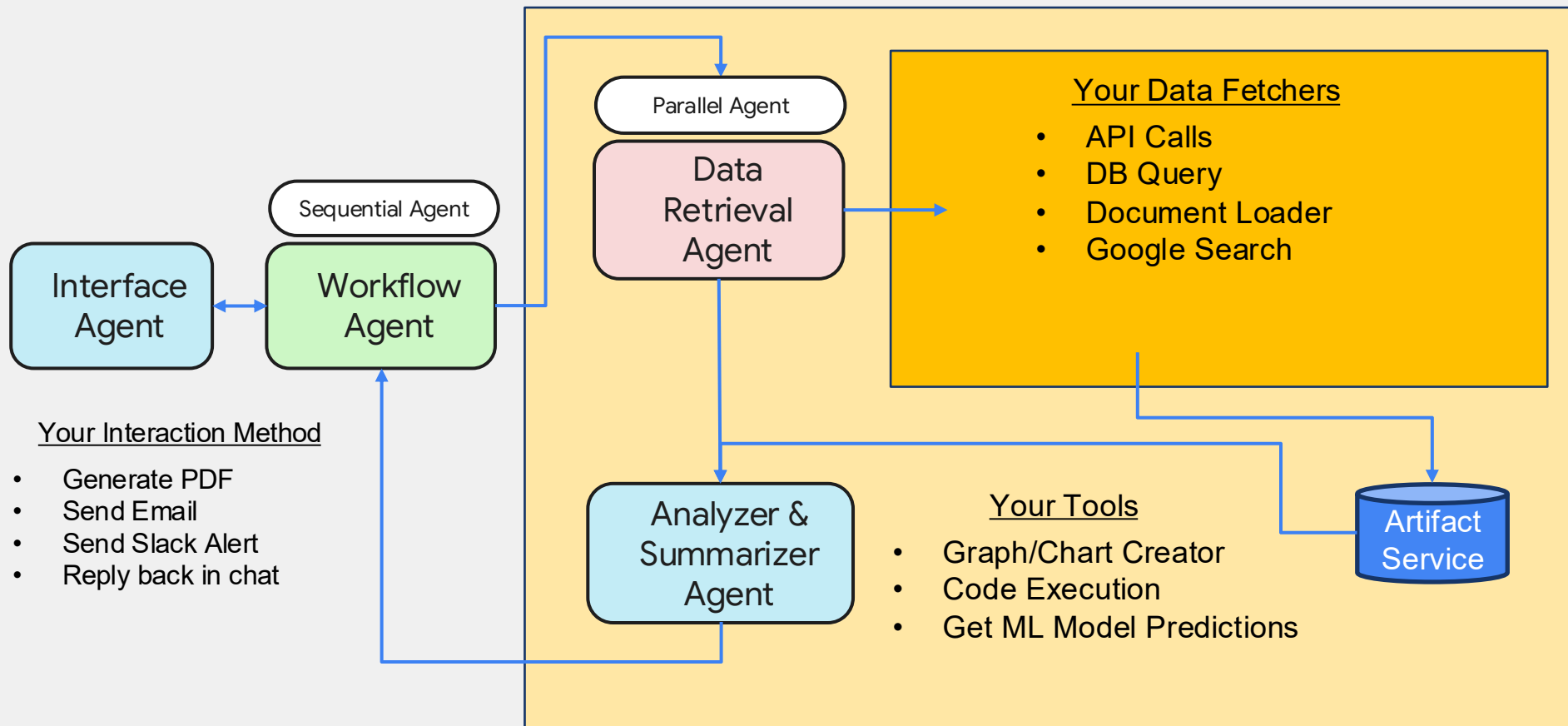
Your Architecture

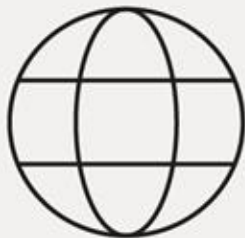


Your Architecture



Your Architecture





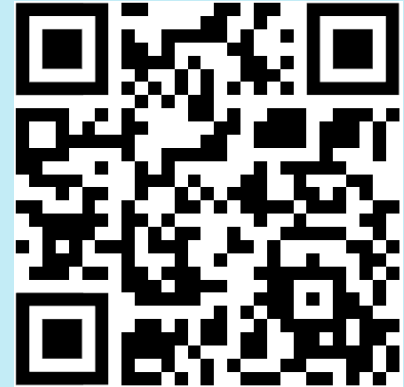
Demo Code on Github



Google
Developer
Groups

Deployment

- Deploy directly to GCP – Cloud Run
- Non-GCP deployments – Docker container
 - Single service
 - Host as ``adk api_server –port 8000``
- Identify good Artifact services
 - RAM – local / low traffic / single deployment
 - GCS good for GCP deployments
 - Redis sidecar for kubernetes deployments
- Serverless deployments
 - Event driven
 - Scalable (to 0 if needed)



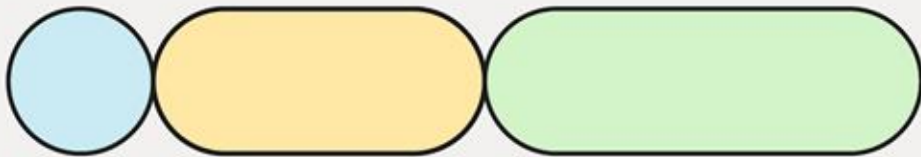
Medium.com Article

Deployment – Gemini Enterprise

- Deploy to Vertex AI Agent Engine
- Link to Gemini Enterprise
 - Deploy, govern, use the agent along with other agents
- Add custom agent
 - Describe agent so LLM can call it if needed
 - Add as tool - use @stockbot to call the agent in chat
- Use A2A protocol to integrate with other agents



Gemini Enterprise docs



Thank You!

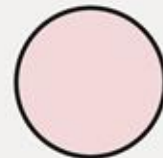
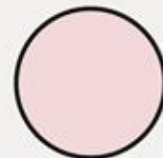
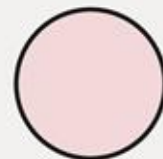


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Google
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